COHOMOLOGY WITH SUPPORT IN A CLOSED SUBSET VANISHES ON A PUSHFORWARD FROM A DISJOINT OPEN SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one cross-chapter label, recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Pushforward of a K-injective complex along an open immersion is K-injective. Stacks Project, Cohomology of Sheaves, Lemma lemma-K-injective-flat (tag 0D5Z). Justifies computing R-H_Z(Rj_*K) from the complex j_*I^bullet, and hence the reduction to abelian sheaves.

Source reference. Original lemma 0D5Z: sources/stacks-chow/tags.txt, source lines 16504–16504. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $Z \subset X$ be a closed subset. Let $j : U \to X$ be the inclusion of an open subset with $U \cap Z = \emptyset$. Then $R\mathcal{H}_Z(Rj_*K) = 0$ for all K in $D(\mathcal{O}_U)$.*

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ of $\mathcal{O}_U$-modules representing K. Then $j_*\mathcal{I}^\bullet$ represents Rj_*K. By Lemma 0D5Z the complex $j_*\mathcal{I}^\bullet$ is a K-injective complex of $\mathcal{O}_X$-modules. Hence $\mathcal{H}_Z(j_*\mathcal{I}^\bullet)$ represents $R\mathcal{H}_Z(Rj_*K)$. Thus it suffices to show that $\mathcal{H}_Z(j_*\mathcal{G}) = 0$ for any abelian sheaf $\mathcal{G}$ on U. Thus we have to show that a section s of $j_*\mathcal{G}$ over some open W which is supported on $W \cap Z$ is zero. The support condition means that $s|_{W \setminus W \cap Z} = 0$. Since $j_*\mathcal{G}(W) = \mathcal{G}(U \cap W) = j_*\mathcal{G}(W \setminus W \cap Z)$ this implies that s is zero as desired. □

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